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Debias Once for All: A Data-Centric Strategy for Fair Machine Learning

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Abstract

The increasing use of deep neural networks (DNNs) in high-stakes domains such as hiring, healthcare, and finance has heightened concerns about algorithmic fairness. Because training data can encode historical and societal biases, learned models may exhibit disparate outcomes for underrepresented groups. Prior work is largely *model-centric*, improving fairness via specialized loss functions or architectural modifications, which can introduce additional training overhead and hinder deployment in modular or rapidly evolving pipelines. We instead study a *data-centric* alternative: constructing a fair training dataset that promotes equitable behavior without changing the model architecture. We propose FairData, which synthesizes a fair dataset by optimizing a gradient-matching objective that aligns the training dynamics of a randomly initialized model on the synthetic data with those on the original data, while explicitly regularizing for group fairness. The resulting dataset is model-agnostic, lightweight, and remains in the original input space, enabling straightforward reuse across downstream models. Experiments on four benchmark datasets show that FairData consistently reduces group disparities across diverse architectures while maintaining competitive predictive performance, suggesting fairness-aware dataset optimization as a practical complement to model-specific fairness techniques.

CCS Concepts

• **Computing methodologies → Machine learning algorithms**;
• **Applied computing → Law, social and behavioral sciences**.

Keywords

Fair Machine Learning, Data-Centric AI, Deep Neural Networks

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1 Introduction

Deep neural networks (DNNs) have become the foundation of modern intelligent systems, finding applications across critical domains such as finance [49], hiring [23], healthcare [21], and education [6, 31]. By learning complex patterns from large-scale datasets, DNNs have revolutionized predictive modeling and decision-making. However, the data-driven nature of these models makes them vulnerable to replicating and even exacerbating societal biases present in their training data [3, 17]. In real-world settings, such biases can lead to disproportionately adverse outcomes for individuals from underrepresented or disadvantaged groups, especially along sensitive attributes such as gender, race, and age [24, 41]. These issues have sparked growing concern among researchers and policymakers, catalyzing a surge of interest in the field of algorithmic fairness [12, 22, 53, 67]. Beyond achieving high accuracy, there is now an increasing demand for AI systems that are transparent, socially responsible, and equitable in their treatment of different subpopulations [47, 48, 55]. Addressing these concerns has led to the development of numerous fairness metrics and mitigation techniques, encompassing diverse perspectives from computer science, law, and ethics.

To address fairness, most efforts have focused on *model-centric approaches*, which can be broadly categorized into *in-processing* and *post-processing* methods. In-processing methods integrate fairness constraints directly into the training objective [5, 29, 63], while post-processing methods adjust model outputs after training to satisfy fairness metrics [22, 25]. While these methods offer flexibility in targeting various fairness notions, they suffer from critical limitations: they are model-dependent, often introduce optimization instability, and lack generalizability across models. For instance, each new model architecture trained on the same dataset may require a separate debiasing process, leading to repetitive tuning efforts and limited reusability of fairness interventions.

In contrast, *data-centric approaches* shift the focus from modifying models to improving the quality of training data itself [46, 59, 65], aligning with the emerging paradigm of Data-Centric AI. Rather than designing increasingly complex architectures, data-centric methods aim to enhance dataset quality to improve both

model performance and fairness. Yet, existing pre-processing techniques, such as reweighting [28], re-labeling [7], and attribute obfuscation, typically require domain-specific prior knowledge and manual heuristics. Moreover, they cannot directly leverage well-established fairness regularizers [9, 45] due to the lack of differentiable optimization objectives over data generation.

Recent advances in *dataset learning* offer a new direction: synthesizing or modifying training datasets by optimizing them to induce desired training behaviors on downstream models [27, 68, 68]. Inspired by the progress in dataset condensation and optimization-based data generation [62], these methods enable the automatic creation of compact or task-tailored datasets that preserve essential information from the original data. However, most existing work in this line focuses solely on optimizing for downstream *accuracy* or data efficiency, largely overlooking critical aspects such as fairness. In high-stakes domains where fairness is a core requirement, models trained on such datasets may still yield discriminatory predictions. This disconnect highlights the need for dataset learning approaches that not only retain utility but also promote fairness during training. In this paper, we ask a fundamental question: ***Can we automatically learn a debiased training dataset that promotes fairness in model predictions, while preserving accuracy and enabling transferability across models***?

To this end, we propose a novel framework called **FairData**, which addresses fairness from a data-centric perspective by directly optimizing a set of *fair data points*. These points are learned such that any model trained on them yields fair and accurate predictions. Our approach builds upon the *gradient matching paradigm* [68], which encourages the synthetic fair data to induce a similar training trajectory (in terms of parameter updates) as the original data, thereby maintaining utility. Simultaneously, we incorporate *differentiable fairness regularizers* [45, 63] into the optimization objective to guide the debiasing process. This design enables our method to optimize for fairness without altering the model structure or relying on label heuristics.

We highlight two core challenges in this setting: **I)**: *How to formulate an objective that balances fairness and utility in data learning*? **II)**: *Can the learned fair data generalize to unseen models and retain cross-architecture fairness*? To address (i), we jointly optimize a loss that combines gradient alignment (for preserving utility) and fairness regularization (for ensuring equity). To address (ii), we investigate the *transferability* of the learned fair data across architectures, showing that our data-centric debiasing generalizes well without retraining on every model. Our contributions are summarized as follows:

- We introduce a new perspective on fairness in deep learning, **data-centric debiasing**, that shifts the focus from models to the dataset, enabling once-for-all debiasing applicable to various architectures.
- We propose FairData, a novel framework that integrates gradient matching and fairness regularization to learn fair datasets. Unlike prior methods, FairData does not alter model architectures or require access to sensitive attributes at test time.
- We evaluate FairData on four real-world datasets across multiple model architectures. The results show improved fairness metrics with competitive accuracy, and the learned dataset transfers well across architectures.

2 Related Work

In this section, we review two major lines of research that inform our proposed approach: algorithmic fairness in machine learning and dataset learning, especially recent advances in data condensation and fairness-aware data selection.

2.1 Algorithmic Fairness in Machine Learning

Ensuring fairness in machine learning has become a crucial challenge in high-impact applications such as hiring, healthcare, criminal justice, and credit scoring, where algorithmic bias may amplify societal inequalities. Theoretical foundations of fairness are generally grouped into group fairness and individual fairness. Group fairness aims for equitable treatment across demographic subgroups, as instantiated by notions such as Demographic Parity (DP), Equalized Odds, and Equality of Opportunity [13, 17, 22, 42, 43]. Individual fairness, on the other hand, mandates that similar individuals should receive similar predictions [17], focusing on the consistency of treatment at a finer granularity.

Based on intervention timing, fairness-enhancing methods can be categorized into three paradigms: (i) Pre-processing methods mitigate bias in the input data distribution before model training [7, 28]. (ii) In-processing techniques incorporate fairness-aware loss functions or constraints into the training process [1, 37–40, 66]. (iii) Post-processing approaches adjust model predictions to meet fairness criteria without retraining [22, 54]. Recent innovations explore more flexible, data-centric strategies that balance fairness with accuracy and interpretability. For example, Falcon introduces a fairness-aware active learning strategy based on multi-armed bandits that prioritizes sample selection using fairness signals [50]. FairSHAP [19] leverages Shapley value attribution to identify fairness-critical training samples and applies minimal perturbations to achieve fairness improvements. Fairpriori [69] combines frequent itemset mining with intersectional fairness auditing, offering a scalable and interpretable way to detect subgroup bias.

Importantly, these approaches increasingly reflect a shift toward model-agnostic, data-aware fairness mechanisms, which do not require modifications to model internals or reliance on protected attribute labels at test time. This shift aligns with our proposed method, which treats data as the primary medium for fairness intervention and optimization.

2.2 Dataset Learning and Condensation

Dataset learning is an emerging paradigm that treats data as a first-class object of optimization. It aims to curate or synthesize datasets that achieve downstream training goals more effectively or efficiently than the original data. Applications include privacy preservation [35, 60], robustness [18], compression [56], and efficiency in distributed or federated learning [14, 70]. A particularly relevant subfield is dataset condensation, which learns small, synthetic datasets that are optimized to mimic the training dynamics of real data. These methods often use gradient matching [68], bi-level optimization [8], or information-aware selection [30, 33] to compress a large dataset into a few informative samples per class. For

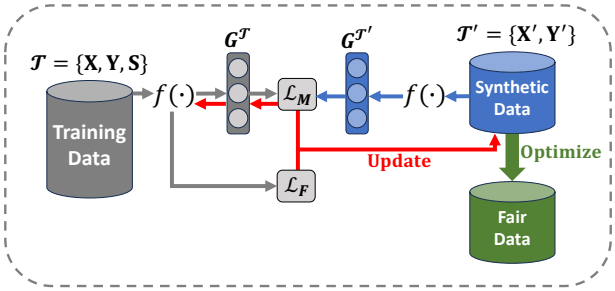


Figure 1: The proposed *method* framework trains the function $f(\cdot)$ on the original dataset $\mathcal{T}$ while capturing the gradient $G^{\mathcal{T}}$. Meantime, it initializes and feeds the synthetic dataset $\mathcal{T}'$ into $f(\cdot)$, obtaining the gradient $G^{\mathcal{T}'}$. Then, two crucial loss components are computed: the gradient matching loss $\mathcal{L}_M$ and the fair loss $\mathcal{L}_F$. These components are then combined to facilitate the update of the synthetic dataset $\mathcal{T}'$.

instance, MTT [8] optimizes synthetic examples by matching gradients across multiple training steps, while IDC [30] uses Jacobian matching to capture feature-level behavior. While effective for improving utility, most dataset condensation methods ignore fairness constraints. Only a few works have started to explore fair dataset synthesis. FairGAN [61] trains generative models with adversarial fairness constraints, but such approaches often suffer from training instability and lack of interpretability. Other works incorporate label smoothing or adversarial fairness penalties [52], but still rely on complex deep generative architectures.

In contrast, our method focuses on learning a fair subset of real data, preserving both the original data structure and interpretability, through optimization over gradient alignment and fairness objectives (see Figure 1). By retaining original features and human-interpretable samples, our approach is particularly suited for applications requiring transparency, auditability, or downstream compatibility with non-differentiable or rule-based models.

3 FairData Framework

In this section, we elaborate on the proposed FairData framework, which integrates two complementary components: (1) *dataset learning* via gradient matching and (2) *data-centric debiasing* via fairness-aware regularization. This design ensures that the learned synthetic dataset not only retains the task-relevant information from the original dataset but also satisfies group fairness criteria.

Recent advances in dataset condensation [33, 34, 68] have demonstrated that small synthetic datasets can emulate the training dynamics of real data when guided by optimization signals such as gradient matching. However, such methods typically ignore fairness concerns and may inherit or amplify biases present in the original data. Motivated by this limitation, our framework aims to co-optimize data fidelity and fairness in a unified training pipeline.

FairData starts by simulating the learning behavior of the original dataset through a gradient matching mechanism, where a synthetic dataset $\mathcal{T}'$ is learned such that the gradients it induces are close to those of the original dataset $\mathcal{T}$ across training epochs. In parallel, a fairness regularization term is imposed on the synthetic data to reduce the disparity of model predictions across different sensitive groups. The joint optimization ensures that the synthetic dataset is both predictive and fair, enabling efficient and bias-mitigated learning.

4 Problem Statement

We study the problem of data-centric debiasing. Given a dataset $\mathcal{T} = \{\mathbf{X}, \mathbf{Y}, \mathbf{S}\}$ with N samples, where $\mathbf{X} \in \mathbb{R}^{N\times D}$ denotes the input features, $\mathbf{Y} \in \{0, \ldots, C-1\}^N$ the class labels, and $\mathbf{S} \in \{0, \ldots, K-1\}^N$ the sensitive attributes, let f_θ be a downstream model parameterized by $\boldsymbol{\theta}$. For any sample $(\mathbf{x}, \mathbf{y}, \mathbf{s}) \in \mathcal{T}$, the model produces a prediction $\hat{\mathbf{y}} = f_\theta(\mathbf{x})$. Our goal is to learn a synthetic yet fair dataset $\mathcal{T}' = \{\mathbf{X}', \mathbf{Y}'\}$, where $\mathbf{X}' \in \mathbb{R}^{N\times D}$ and $\mathbf{Y}' \in \{0, \ldots, C-1\}^N$, such that **(i)** a model trained on $\mathcal{T}'$ achieves competitive predictive performance to one trained on the original dataset $\mathcal{T}$, and **(ii)** the resulting predictions $\hat{\mathbf{y}}' = f_\theta(\mathbf{x}')$ exhibit improved group fairness (e.g., demographic parity or equalized odds) with respect to the sensitive attributes.

4.1 Dataset Learning via Gradient Matching

Enlightened from previous studies [26, 27, 68], we approach by gradient matching, which optimizes a synthetic dataset by minimizing the total gradient distances of the synthetic and the original datasets in training. At a specific epoch t in training, given the classification losses for $\mathcal{T}$ and $\mathcal{T}'$, denote as $\mathcal{L}_t^{\mathcal{T}}$ and $\mathcal{L}_t^{\mathcal{T}'}$, we calculate the gradient matrices $D_t^{\mathcal{T}}$ and $D_t^{\mathcal{T}'}$ for these two datasets as:

$$\begin{aligned}
\mathcal{L}_t^{\tilde{\mathcal{T}}} &= \frac{1}{|\tilde{\mathcal{T}}|} \sum_{(\mathbf{x},\mathbf{y})\in\tilde{\mathcal{T}}} \ell(\mathrm{DNN}_{\theta^t_{\tilde{\mathcal{T}}}}(\mathbf{x}), \mathbf{y}), \\
\mathcal{L}_t^{\mathcal{T}'} &= \frac{1}{|\mathcal{T}'|} \sum_{(\mathbf{x}'_t,\mathbf{y}'_t)\in\mathcal{T}'} \ell(\mathrm{DNN}_{\theta^t_{\mathcal{T}'}}(\mathbf{x}'_t), \mathbf{y}'_t), \\
D_t^{\mathcal{T}} &= \nabla_{\theta^t_{\mathcal{T}}} \mathcal{L}_t^{\mathcal{T}}(\boldsymbol{\theta}^t_{\mathcal{T}}), \quad D_t^{\mathcal{T}'} = \nabla_{\theta^t_{\mathcal{T}'}} \mathcal{L}_t^{\mathcal{T}'}(\boldsymbol{\theta}^t_{\mathcal{T}'})
\end{aligned} \tag{1}$$

where $\boldsymbol{\theta}^t_{\mathcal{T}}$ and $\boldsymbol{\theta}^t_{\mathcal{T}'}$ are the model parameters trained on the original dataset $\mathcal{T}$ and the learned dataset $\mathcal{T}'$ at epoch t. The idea of the dataset learning is to minimize the sum of the gradient matching loss $\mathcal{L}_M$ of each epoch.

Gradient Matching. To ensure task consistency, we adopt a gradient matching objective [68], which minimizes the discrepancy between the gradients computed from the original dataset $\mathcal{T}$ and the synthetic dataset $\mathcal{T}'$ at each training step t:

$$\min_{\mathcal{T}'} \mathbb{E}_{\theta^0\sim P_{\theta^0}} \left[\sum_{t=0}^{T-1} \mathcal{L}_M(D_t^{\mathcal{T}}, D_t^{\mathcal{T}'}) \right], \tag{2}$$

where $D_t^{\mathcal{T}}$ and $D_t^{\mathcal{T}'}$ denote the gradients of model parameters on $\mathcal{T}$ and $\mathcal{T}'$.

Gradient Distance Metric. We adopt the cosine distance to measure directional alignment between gradients:

$$\mathcal{L}_M(D^{\mathcal{T}}, D^{\mathcal{T}'}) = \sum_{c=0}^{C-1} \sum_{l=1}^{L} \sum_{b=1}^{B} \left(1 - \frac{D_{l,b}^{\mathcal{T}_c} \cdot D_{l,b}^{\mathcal{T}'_c}}{\left\| D_{l,b}^{\mathcal{T}_c} \right\| \left\| D_{l,b}^{\mathcal{T}'_c} \right\|} \right), \tag{3}$$

which aggregates the cosine distance of class-wise gradients across all layers and parameters. where $D_t^{\mathcal{T}}$ and $D_t^{\mathcal{T}'}$ denote the gradients computed on the original and synthetic datasets at step t, respectively. This formulation emphasizes the alignment of gradient directions, which is crucial for capturing optimization trajectories.

4.2 Data-centric Debiasing

While gradient matching ensures that the synthetic dataset $\mathcal{T}' = \{\mathbf{X}', \mathbf{Y}'\}$ retains the utility of the original dataset $\mathcal{T} = \{\mathbf{X}, \mathbf{Y}, \mathbf{S}\}$, it does not guarantee fairness. In data-centric machine learning, the quality and structure of the data itself play a crucial role in downstream performance, including fairness. Thus, instead of solely relying on fairness-aware model design, we explicitly incorporate fairness regularization into the data synthesis process. This aligns with our overall data-centric objective: to learn a *condensed yet fair* dataset that preserves both accuracy and fairness without retraining large models.

To operationalize fairness, we incorporate a regularization term into the dataset optimization objective that penalizes group disparities in predicted outcomes. Specifically, we adopt the *demographic parity gap* Δ_{DP} [11, 64], which measures the deviation in label distributions across sensitive groups. This is defined as:

$$\mathcal{L}_F = \frac{1}{K} \sum_{k=0}^{K-1} \max_c |P(\hat{\mathbf{y}}' = c) - P(\hat{\mathbf{y}}' = c | \mathbf{s} = k)|, \tag{4}$$

where $\hat{\mathbf{y}}' = f_\theta(\mathbf{x}')$ is the model prediction on $\mathcal{T}'$, and K is the number of sensitive groups.

This loss term is differentiable with respect to both the model parameters $\boldsymbol{\theta}$ and the synthetic data $\mathbf{X}'$, enabling direct optimization during condensation. By minimizing $\mathcal{L}_F$, we encourage the model trained on $\mathcal{T}'$ to produce similar predictions across all subgroups, thus reducing bias propagated from imbalanced or unrepresentative training samples. The final training objective integrates utility, fairness, and regularization:

$$\mathcal{L} = \mathcal{L}_F + \alpha \mathcal{L}_M + \beta \|\boldsymbol{\theta}\|^2, \tag{5}$$

where α, β are hyperparameters. This loss promotes fairness while preserving the learning signal. We initialize $\mathbf{Y}'$ by sampling from the class distribution of $\mathbf{Y}$ to stabilize training. The final objective and the updated rules can be formulated as:

$$\begin{aligned} &\min_{\mathbf{X}', \boldsymbol{\theta}} \mathrm{E}_{\boldsymbol{\theta}^0 \sim P_{\boldsymbol{\theta}^0}} \left[\sum_{t=0}^{T-1} \mathcal{L}_M \left(D_t^{\mathcal{T}}, D_t^{\mathcal{T}'} \right) \right], \\ &\text{with } \mathbf{X}'_{t+1} = \mathbf{X}'_t - \eta_1 \nabla_{\mathbf{X}'_t} \mathcal{L}_M, \\ &\text{and } \boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t - \eta_2 \nabla_{\boldsymbol{\theta}^t_{\mathcal{T}'}} \mathcal{L} \left(f_{\boldsymbol{\theta}^t}(\mathbf{X}'), \mathbf{Y}' \right), \end{aligned} \tag{6}$$

where $\boldsymbol{\theta}^t$ can use the model parameter trained on the synthetic dataset, i.e., $\boldsymbol{\theta}^t_{\mathcal{T}'}$.

4.3 Optimization

Following the construction of the condensed dataset $\mathcal{T}' = \{\mathbf{X}', \mathbf{Y}'\}$ and the integration of fairness-aware objectives, we now describe the optimization process that enables $\mathcal{T}'$ to serve as a compact and fair proxy for training models.

Label Treatment. Each synthetic feature vector $\mathbf{x}'_i \in \mathbf{X}'$ is associated with a fixed label $y'_i \in \mathbf{Y}'$ throughout optimization. This static labeling strategy, widely adopted in dataset condensation [68], simplifies the learning dynamics and ensures stability. Moreover, in fairness-sensitive tasks, preserving label semantics is critical for ensuring meaningful inter-group comparisons and for interpreting how synthetic samples affect group-level outcomes.

Algorithm 1: FairData: Fair Dataset Condensation

```
Input: T = (X,Y,S), initialized Y'
Randomly initialize X' and fix Y'
for j = 0, ..., J − 1 do
    Initialize θ^0 ~ P_θ^0
    for t = 0, ..., T − 1 do
        Compute D_t^T, D_t^T'
        Compute L_M, L_F, L
        if t%(t_1 + t_2) < t_1 then
            Update X'_{t+1} = X'_t − η_1 ∇_{X'_t} L_M
        else
            Update θ^{t+1} = θ^t − η_2 ∇_{θ^t} L
Return: T' = {X'_{T−1}, Y'}
```

Alternating Update Strategy. Directly optimizing both $\mathbf{X}'$ (the learned dataset) and model weights $\boldsymbol{\theta}$ is non-trivial due to their tight interdependence. Specifically, updating $\mathbf{X}'$ influences the model gradients, which in turn affect how $\boldsymbol{\theta}$ is optimized; this forms a bilevel optimization problem that is computationally expensive and difficult to solve jointly. To address this challenge, we adopt an alternating update scheme that decouples the optimization of $\mathbf{X}'$ and $\boldsymbol{\theta}$. At each iteration, we first fix $\boldsymbol{\theta}$ and optimize $\mathbf{X}'$ to minimize the matching loss $\mathcal{L}_M$ for t_1 steps. This encourages the synthetic data $\mathbf{X}'$ to approximate the gradient behavior of real data. Next, we fix the updated $\mathbf{X}'$ and update $\boldsymbol{\theta}$ by minimizing the standard training loss $\mathcal{L}$ for t_2 steps, effectively training the model on the current version of the learned dataset. This alternating strategy has several advantages: (1) it reduces the complexity of the joint optimization by breaking it into two simpler subproblems, (2) it ensures that the learned data remains aligned with the evolving model dynamics, and (3) it allows stable and efficient convergence in practice. We repeat this process until convergence, typically requiring only a few epochs due to the compact size of $\mathbf{X}'$.

Fairness Consideration. Although the sensitive attribute $\mathbf{S}$ is not explicitly encoded in $\mathcal{T}'$, our optimization process indirectly incorporates fairness constraints. This is achieved by evaluating model fairness on the original dataset $\mathcal{T}$ during training and penalizing group-level disparities in predictions (see Section 4.2). In addition, the use of gradient matching aligns parameter updates across groups, embedding an implicit fairness inductive bias. By jointly optimizing utility and fairness objectives, we encourage the condensed dataset to generalize fairly across demographic groups without directly memorizing sensitive attributes.

4.4 Discussion

FairData offers several benefits over traditional fairness-aware training: (1) It decouples debiasing from full retraining by operating purely on data; (2) It significantly reduces data storage and training costs through condensation; and (3) It is compatible with

Table 1: Detailed statistics of the datasets used in our experiments. The table reports the number of data samples, input features, prediction targets, sensitive attributes, and the data type for each dataset.

Dataset	Adult	COMPAS	ACS-I	CelebA-A
# Data	45, 222	6, 172	195, 655	202, 599
# Feat.	101	405	908	48×48
# Target	Income	Credit	Income	Attractive
# Sens.	Gender, Race	Gender, Race	Gender, Race	Gender, Age
Type	Tabular	Tabular	Tabular	Image

various fairness definitions beyond demographic parity, such as equal opportunity [22], by modifying $\mathcal{L}_F$. Thus, FairData paves a promising path for efficient and fair data-centric learning.

5 Experiments

In this section, we first evaluate the effectiveness of the proposed method in Section 5.3 across three tabular datasets and one image dataset, comparing with six debiasing baselines. Then we assess the accuracy-fairness trade-offs in Section 5.4. In addition, we conduct a distribution visualization of the fair data generated through our approach in Section 5.6. Finally, we investigate the impact of the control parameter discussed in Section 5.7 to gain deeper insights into its influence on the overall performance of our method.

5.1 Datasets and Baselines

To comprehensively assess the effectiveness of FairData, we employ diverse benchmark datasets covering both tabular and image, and compare against representative baselines spanning standard training, in-processing, and data-centric debiasing techniques.

Datasets. We evaluate the performance of FairData across four widely adopted fairness benchmarks. The Adult dataset [4], derived from the 1994 U.S. Census, targets income classification (> $50,000) based on demographic and employment features. The COMPAS dataset [32] contains criminal defendant records and is used to predict the likelihood of recidivism within two years. The ACS-I dataset [16] offers a modern variant of census-based prediction tasks focused on income thresholds. Lastly, CelebA-A [44] consists of 20,000 celebrity facial images, where the task is to classify visual attributes, such as "smiling", while ensuring demographic fairness across sensitive groups. Dataset statistics are provided in Table 1.

Baselines. We compare FairData against a representative set of baselines, covering both standard training and fairness-aware methods. **(i)** ERM is the conventional Empirical Risk Minimization approach, which solely optimizes for predictive performance without accounting for fairness. **(ii)** DiffDP [11] introduces a regularization term that minimizes the demographic parity gap during training. **(iii)** HSIC minimizes statistical dependence between model outputs and sensitive attributes using the Hilbert–Schmidt Independence Criterion. **(iv)** PRemover [29] explicitly reduces mutual information between predictions and sensitive features. **(v)** AdvDebias [5, 63] adopts an adversarial training framework that discourages the model from encoding sensitive information by jointly optimizing with a discriminator. **(vi)** LAFTR [45] learns fair representations by jointly minimizing classification error, reconstruction loss, and the adversary's ability to predict sensitive attributes. **(vii)** OptimPrep [7] is a data-centric pre-processing method that learns a mapping to transform the training set into a fairer version, without altering the downstream model.

5.2 Evaluation Metrics and Setups

Evaluation Metrics. To evaluate model performance, we report both accuracy and AUC (Area Under the Receiver Operating Characteristic Curve). In terms of fairness, we adopt two widely used group fairness metrics. *Demographic Parity (DP)* [17] quantifies the difference in prediction distributions across sensitive groups, aiming for statistical independence between the predicted label $\hat{\mathbf{Y}}$ and sensitive attribute $\mathbf{S}$. *Equalized Odds (EO)* [22] requires that predictions be conditionally independent of the sensitive attribute given the true label, thereby equalizing true and false positive rates across groups. Lower values of both metrics indicate stronger fairness guarantees.

Experimental Setup. For tabular datasets, we employ a two-layer multilayer perceptron (MLP) with 256 hidden units and ReLU activations. For image-based tasks, we use a ResNet-18 backbone pre-trained on ImageNet and fine-tuned on the downstream classification objective. All models are trained using the Adam optimizer [15] with a learning rate of 0.01 for 150 optimization steps. The batch size is set to 4096 for tabular datasets and 128 for image datasets. Unless otherwise noted, we use hyperparameters $\alpha = 1.4$, $\beta = 10^{-3}$ for tabular tasks and $\alpha = 2.5$, $\beta = 10^{-2}$ for image tasks. All results are averaged across three independent runs with different random seeds to ensure statistical robustness.

5.3 Effectiveness of FairData

To assess the overall effectiveness of FairData, we conduct a thorough comparison with a broad set of fairness baselines, including both pre-processing and in-processing methods, across four benchmark datasets: Adult, COMPAS, ACS-I, and CelebA-A. These datasets span multiple modalities (tabular and image) and sensitive attributes (gender, race, age), making them ideal testbeds to evaluate generalizability. We report results using standard metrics for utility (Accuracy, AUC) and fairness (Δ_{EO}, Δ_{DP}), averaged over 10 runs as shown in Table 2.

Overall, **FairData achieves the best or competitive performance across all datasets, consistently outperforming existing baselines in both fairness and accuracy**. This validates the robustness, adaptability, and plug-and-play nature of our method.

- **Superior Fairness-Utility Trade-off.** Unlike in-processing baselines like AdvDebias and HSIC, which enforce fairness constraints during model training, FairData decouples fairness from model architecture and instead learns fair data directly. As a result, it achieves better fairness with less compromise in accuracy. For instance, on Adult (Gender), FairData attains the **lowest Δ_{DP} of 0.5±1.1** and the **highest accuracy of 84.2±0.9**, outperforming all baselines across both objectives.
- **Stronger Performance than Pre-processing Approaches.** Compared to traditional methods like OptimPrep, FairData offers finer-grained control and more stable performance. Its gradient matching framework aligns the debiased data distribution with the training dynamics of the target task, rather than relying on coarse heuristics. This is evident in datasets like COMPAS (Race),

Table 2: Accuracy and fairness results of our method (FairData) and existing baselines across four datasets and multiple sensitive attributes. We report the average and standard deviation over 10 runs. The best results are shown in bold. ↑ indicates higher values are better (e.g., accuracy, AUC), while ↓ indicates lower values are better (e.g., fairness gaps).

Dataset	Sens.	Metric	ERM	DiffDP	HSIC	AdvDebias	LAFTR	PRemover	OptimPrep	FairData
Adult	Race	Acc. (↑)	85.2±0.3	83.1±1.9	84.4±0.5	80.6±4.4	83.0±3.2	79.1±3.5	82.9±2.1	**84.7±0.8**
		AUC (↑)	91.1±0.3	88.5±1.1	89.9±0.7	88.3±7.9	88.5±7.4	87.1±0.9	89.6±0.1	**90.1±0.7**
		Δ_{EO} (↓)	13.1±2.7	3.4±2.3	3.9±3.1	5.6±5.8	11.6±7.7	2.5±2.4	10.9±2.3	**2.4±1.2**
		Δ_{DP} (↓)	12.2±0.7	5.3±2.2	6.5±2.0	5.5±4.9	9.3±4.0	2.7±2.5	8.6±1.2	**2.1±1.0**
	Gender	Acc. (↑)	85.4±0.3	82.7±1.1	82.4±0.7	82.5±2.3	83.8±2.2	77.5±3.0	83.1±0.7	**84.2±0.9**
		AUC (↑)	90.8±0.2	88.0±1.5	86.6±1.0	86.1±8.0	90.4±0.8	82.8±2.6	82.8±2.6	**90.6±0.6**
		Δ_{EO} (↓)	14.2±3.2	36.0±10.7	33.6±5.9	20.4±19.4	11.3±9.0	10.0±14.6	13.0±3.5	**5.3±2.1**
		Δ_{DP} (↓)	16.7±0.7	3.2±3.0	0.9±0.7	14.1±8.5	11.5±4.4	0.5±0.6	12.5±5.8	**0.5±1.1**
COMPAS	Race	Acc. (↑)	73.0±0.6	59.8±5.3	57.1±3.6	65.4±2.8	66.0±2.0	66.1±1.8	66.5±2.1	**67.3±2.5**
		AUC (↑)	72.6±0.7	68.2±5.3	69.4±2.7	69.5±7.3	71.9±1.4	72.4±1.0	71.3±1.9	**72.8±1.5**
		Δ_{EO} (↓)	29.0±6.7	11.8±12.3	6.1±8.0	27.0±8.4	26.3±7.1	25.2±5.9	25.6±2.4	**4.2±4.7**
		Δ_{DP} (↓)	16.8±3.5	6.6±6.8	3.2±4.3	15.5±4.3	14.9±3.8	14.4±3.1	16.4±5.3	**2.3±3.5**
	Gender	Acc. (↑)	67.1±0.8	61.4±4.2	57.3±2.0	63.3±4.9	63.9±4.6	65.5±2.2	65.2±2.7	**66.6±2.8**
		AUC (↑)	72.1±0.9	68.8±4.5	70.3±2.4	68.8±6.5	70.3±3.8	71.5±0.7	70.5±2.3	**71.8±1.2**
		Δ_{EO} (↓)	19.7±6.0	9.8±8.5	12.1±6.8	16.3±9.2	16.9±9.3	18.6±4.7	17.5±5.5	**5.2±3.4**
		Δ_{DP} (↓)	13.4±2.5	6.1±5.5	7.6±4.3	10.6±5.5	10.7±5.4	12.1±2.8	12.4±4.3	**3.3±1.0**
ACS-I	Race	Acc. (↑)	90.4±0.1	80.2±1.2	81.2±0.4	80.0±6.9	82.1±0.4	74.2±7.3	82.1±2.0	**83.0±3.7**
		AUC (↑)	90.2±0.1	89.5±0.4	89.5±0.3	88.0±6.7	**90.1±0.4**	84.8±2.1	88.6±1.8	89.7±2.1
		Δ_{EO} (↓)	9.7±0.7	6.3±1.9	7.5±2.3	9.5±6.0	7.3±1.5	7.1±4.8	8.3±1.7	**3.8±2.3**
		Δ_{DP} (↓)	9.8±0.4	2.0±1.5	1.3±1.1	9.0±3.6	8.6±0.8	1.0±1.4	8.2±2.2	**0.8±1.0**
	Gender	Acc. (↑)	82.3±0.1	81.2±0.3	76.3±2.0	78.8±9.3	**82.1±0.3**	73.6±8.6	82.0±2.4	82.0±0.7
		AUC (↑)	90.1±0.1	89.1±0.5	89.3±0.3	68.6±9.2	**90.1±0.3**	85.4±2.6	88.4±1.6	89.8±0.4
		Δ_{EO} (↓)	6.2±0.6	12.8±1.5	13.6±1.2	6.6±6.7	2.1±0.9	9.1±5.7	6.3±3.2	**1.6±0.8**
		Δ_{DP} (↓)	9.1±0.3	1.2±0.9	0.9±0.7	5.7±4.2	7.8±1.0	0.6±0.7	7.4±3.9	**0.6±2.2**
CelebA-A	Age	Acc. (↑)	78.2±0.4	67.3±9.1	64.6±4.5	66.5±7.8	68.4±6.1	63.2±5.6	68.3±2.5	**69.1±3.7**
		AUC (↑)	86.7±0.5	74.7±8.6	71.2±4.6	75.3±0.9	76.3±2.7	69.9±5.8	76.1±1.8	**77.3±2.1**
		Δ_{EO} (↓)	37.4±2.3	13.7±11.8	15.2±6.4	27.4±11.2	10.9±6.2	12.7±6.5	4.3±2.7	**5.8±4.3**
		Δ_{DP} (↓)	41.9±1.0	23.7±15.4	4.9±3.7	11.2±21.4	7.8±7.3	5.0±4.0	20.5±13.6	**2.5±3.8**
	Gender	Acc. (↑)	78.2±0.4	66.5±9.2	59.2±5.1	67.2±1.9	67.9±10.3	59.2±5.2	67.0±1.7	**68.1±2.6**
		AUC (↑)	86.7±0.5	73.6±7.5	64.3±5.3	74.2±2.1	73.2±5.2	64.5±5.5	74.3±1.1	**75.6±3.6**
		Δ_{EO} (↓)	70.8±3.2	20.1±13.3	9.7±6.0	17.9±11.3	14.3±8.2	8.0±5.4	21.5±7.4	**6.3±7.9**
		Δ_{DP} (↓)	52.4±1.3	24.2±7.3	4.9±3.2	25.6±19.3	8.9±5.6	6.1±3.7	26.9±6.3	**3.7±2.4**

where FairData achieves the highest AUC of **73.1±1.5** while also reducing Δ_{DP} to **2.3±3.5**.

- **Robustness Across Modalities and Sensitive Attributes.** Fairness-aware methods often perform poorly on complex high-dimensional data such as images. However, FairData maintains its effectiveness even on CelebA-A. For example, on CelebA-A (Gender), FairData achieves a competitive accuracy of **68.1±2.6** and a low Δ_{DP} of **3.7±2.4**, outperforming all other baselines.
- **Low Variance and High Stability.** Across multiple runs, we observe that FairData exhibits lower standard deviations in both utility and fairness metrics compared to prior work. This demonstrates its stability and consistency in optimization, especially in imbalanced or noisy settings.

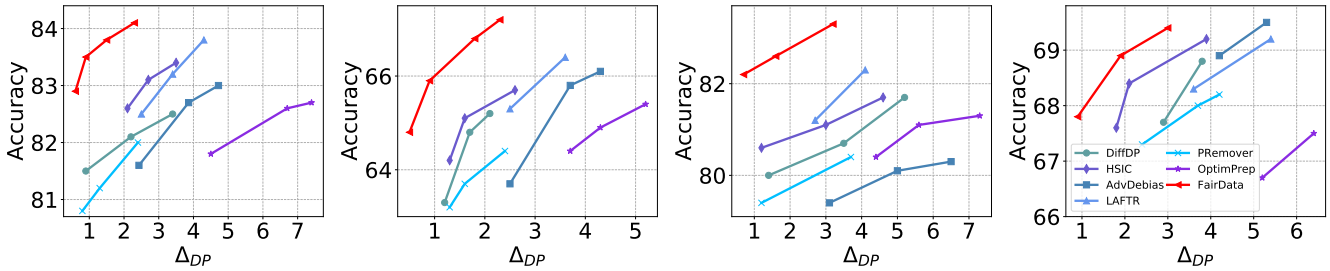
In summary, **FairData combines high performance, fairness, and model-agnostic applicability**, making it a strong candidate for practical deployment.

5.4 Accuracy-Fairness Trade-off

To further investigate how well each method balances prediction performance and fairness, we conduct a Pareto analysis by sweeping key hyperparameters and plotting the resulting trade-offs between accuracy and fairness (measured by Δ_{DP}). Following prior work [36], we visualize the accuracy–fairness frontier for each method across all datasets in Figure 2. A point closer to the upper-left corner represents a more desirable trade-off, achieving higher accuracy and lower bias.

Across the evaluated datasets, FairData frequently lies on or near the Pareto frontier, suggesting a favorable balance between accuracy and fairness. We attribute this behavior to the combination of (i) gradient matching, which helps preserve utility-relevant training dynamics, and (ii) an explicit fairness regularizer that encourages reduced group disparities during dataset optimization. In contrast to prior approaches that may trade substantial accuracy for debiasing (or vice versa), FairData achieves competitive predictive performance while improving fairness metrics, often with a limited compromise in utility.

- Compared to in-processing methods like HSIC and LAFTR, which integrate fairness into the training objective, FairData shows better Pareto efficiency, especially on CelebA-A and Adult, where



(a) Adult (b) COMPAS (c) ACS-I (d) CelebA-A

Figure 2: Pareto frontier of accuracy vs. fairness (Δ_{DP}) across four datasets. Results located in the upper-left corner are preferable. FairData consistently achieves better trade-offs.

Table 3: Transferability performance of FairData across different architectures. Top: `CelebA-A` (age); Bottom: `Adult` (race). Each cell reports the performance of a downstream model (`Te`) trained on fair data generated by a different model (`Tr`). Diagonal entries represent training and testing on the same model. Gray cells indicate that the performance is within a small margin of the corresponding column's diagonal value, specifically, within 1.5 for Accuracy and 0.5 for Δ_{DP}, showing effective transferability from the training model to the test model. MLP denotes a two-layer MLP with 128 hidden units, and MLP-Deep is a three-layer variant with 256 units.

(a) Accuracy on `CelebA-A` (age).

Te\Tr	MLP	ConvNet	ResNet	AlexNet	VGG
MLP	66.4±0.9	67.1±2.0	66.3±1.6	66.1±1.5	65.6±1.8
ConvNet	64.5±1.6	68.5±2.4	68.9±1.8	66.9±1.4	66.4±1.3
ResNet	63.5±2.7	66.2±1.3	69.1±2.2	67.6±1.5	67.2±2.6
AlexNet	65.8±1.9	67.3±3.1	67.0±1.6	67.4±3.7	66.2±2.5
VGG	60.5±2.8	68.6±2.2	67.5±1.5	66.2±2.0	67.7±2.4

(b) Δ_{DP} on `CelebA-A` (age).

Te\Tr	MLP	ConvNet	ResNet	AlexNet	VGG
MLP	7.4±5.1	9.2±4.5	13.2±7.3	11.5±8.8	8.4±3.2
ConvNet	8.2±3.4	5.3±6.3	6.6±4.7	12.4±5.2	6.5±2.0
ResNet	7.1±4.5	8.6±5.3	3.8±1.2	4.6±4.6	7.2±4.6
AlexNet	11.2±6.8	12.7±8.9	10.5±6.3	6.7±4.9	9.9±6.3
VGG	10.3±7.0	14.9±4.1	12.3±8.4	8.5±5.1	5.7±3.8

(c) Accuracy on `Adult` (race).

Te\Tr	MLP	TabNet	FT-Trans	XGBoost	MLP-Deep
MLP	88.4±0.7	83.3±1.2	82.6±1.0	84.0±0.9	85.2±0.8
TabNet	84.6±0.8	87.3±1.1	83.7±1.0	84.5±0.7	83.2±0.6
FT-Trans	82.9±1.0	83.5±0.9	87.1±0.8	83.4±1.1	82.8±1.0
XGBoost	85.3±0.9	84.0±0.7	83.6±1.2	88.0±0.6	84.4±0.9
MLP-Deep	86.2±0.6	83.9±0.8	82.2±1.1	84.1±0.7	88.5±0.7

(d) Δ_{DP} on `Adult` (race).

Te\Tr	MLP	TabNet	FT-Trans	XGBoost	MLP-Deep
MLP	1.6±0.8	1.9±0.8	2.7±0.7	2.2±0.8	2.4±0.6
TabNet	2.2±0.7	2.4±0.6	2.5±0.3	2.0±0.4	2.9±0.3
FT-Trans	1.9±0.7	2.6±0.5	0.5±0.7	1.8±0.9	2.6±0.5
XGBoost	2.9±0.6	2.6±0.8	2.4±0.5	1.3±0.4	1.8±0.5
MLP-Deep	2.1±0.4	2.3±1.0	3.0±0.9	2.7±0.7	0.6±0.9

it achieves higher accuracy at similar or lower Δ_{DP}. For instance, on `CelebA-A` (Gender), FairData reaches 68.1% accuracy with $\Delta_{DP} = 3.7$, outperforming others in both metrics.

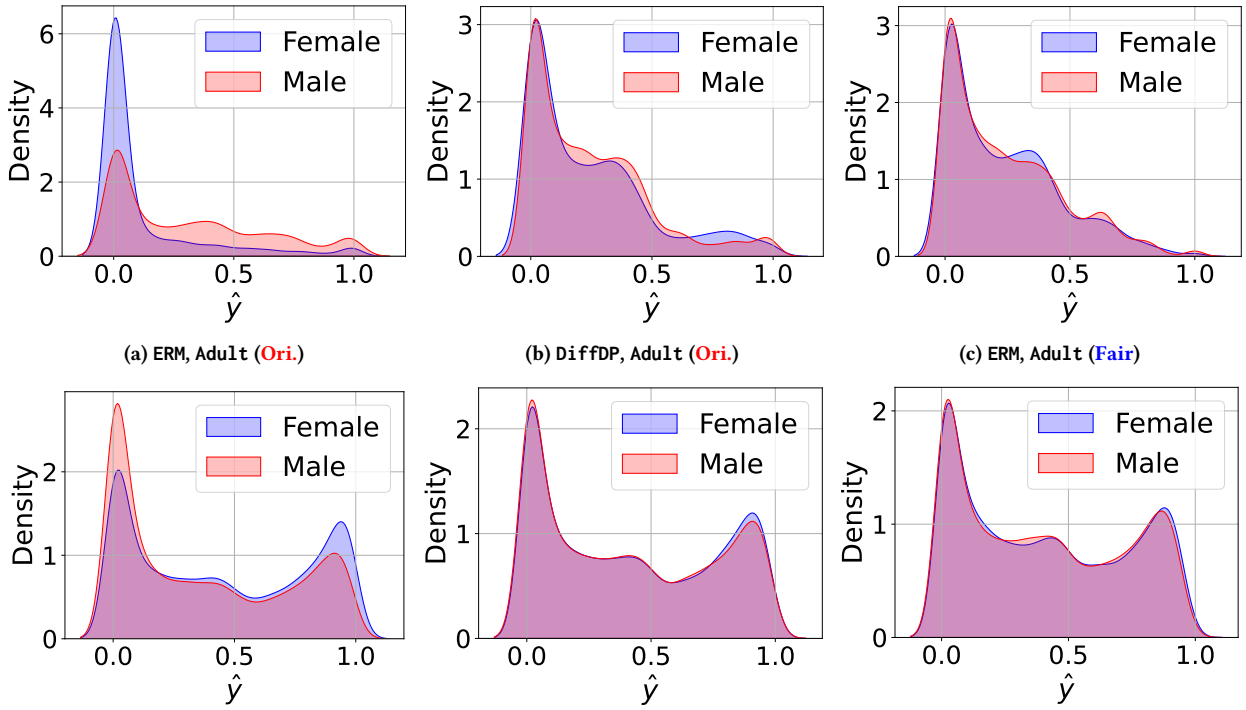
- Pre-processing methods such as `OptimPrep` generally yield sub-optimal results across all datasets, as they are agnostic to downstream tasks. Meanwhile, adversarial methods like `AdvDebias` show greater performance fluctuations, particularly on `Adult` and `COMPAS`, due to optimization instability.
- Overall, the results suggest that FairData provides a favorable utility-fairness trade-off across both tabular and image datasets, and across sensitive attributes including Gender, Race, and Age.

5.5 Transferability of FairData

To assess the generalizability of FairData across model architectures, we conduct a transferability analysis wherein the fair data is generated using one architecture and then consumed by another for downstream training. This simulates real-world deployment scenarios where the architecture used for training may differ from that used at inference due to hardware limitations, deployment pipelines, or model upgrades. We consider two datasets: `CelebA-A` (age), evaluated with five vision backbones (MLP, ConvNet, ResNet, AlexNet, VGG), and `Adult` (race), evaluated with five tabular learners (MLP, TabNet [2], FT-Transformer [20], XGBoost [10], and a deeper variant MLP-Deep).

To ensure fair comparison and isolate architecture-specific factors, we normalize model capacities: TabNet is configured with 2 decision steps and 64 hidden units, FT-Transformer uses 1 block with 2 heads and 128-dimensional hidden states, and XGBoost employs 100 trees with max depth 3. MLP denotes a two-layer model with 256 units, while MLP-Deep has three layers.

Accuracy Transfer. Results in Table 3 (a,c) demonstrate that FairData exhibits strong accuracy transfer across architectures. In `CelebA-A`, most off-diagonal entries are within 1.5% of the diagonal performance—e.g., ConvNet-generated data achieves 68.9% accuracy on ResNet and 66.9% on AlexNet, nearly matching its own 68.5%. Similarly, in `Adult`, TabNet-generated fair data yields 84.5%



(a) ERM, Adult (Ori.) (b) DiffDP, Adult (Ori.) (c) ERM, Adult (Fair)

(d) ERM, ACS-I (Ori.) (e) DiffDP, ACS-I (Ori.) (f) ERM, ACS-I (Fair)

Figure 3: Density distributions of predicted probabilities $\hat{y}$ across gender groups on the Adult and ACS-I datasets. Ori. and Fair indicate the original data and the fair data learned via FairData. Models trained on fair data yield more aligned distributions between groups, indicating improved fairness.

accuracy when used with XGBoost, compared to its self-model performance of 87.3%. This indicates that the learned fair data is not over-specialized to the training model's inductive biases, but rather captures a transferable structure usable by diverse learners.

Fairness Transfer. We further evaluate fairness transfer using the demographic parity gap Δ_{DP} in Table 3 (b,d). Across both datasets, we observe that FairData preserves fairness even when test-time architectures differ from the generator model. For instance, in CelebA-A, fair data from ResNet achieves $\Delta_{DP} = 4.6$ with AlexNet, lower than ResNet's own score of 7.1. Likewise, data from VGG yields $\Delta_{DP} = 8.4$ on MLP, again improving upon its self-trained counterpart. In Adult, the most transferable fairness is observed from MLP-Deep: all downstream models trained on its data achieve $\Delta_{DP} < 3$, with MLP-Deep→MLP yielding a strong Δ_{DP} of just 2.4.

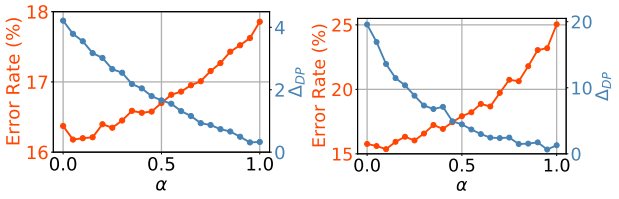
Interpretation. These findings confirm that FairData does not rely on architecture-specific constraints or modeling assumptions. Instead, it distills fairness signals into the data distribution itself by aligning gradient trajectories during generation. This model-agnostic fairness transfer is critical for deployment flexibility, allowing a single fair dataset to support multiple downstream consumers, including shallow models or non-neural learners.

5.6 Distribution Visualization

To better understand the fairness properties of the data produced by FairData, we conduct a distributional analysis of predicted probabilities $\hat{y}$ with respect to the sensitive attribute *gender* on the Adult and ACS-I datasets. This experiment complements previous fairness metrics by providing a more intuitive, visual inspection of model behavior across demographic groups. Specifically, we aim to evaluate whether the predictive distributions are aligned between gender groups, which reflects statistical parity in decision-making. The visualization results are shown in Figure 3.

We compare three settings for each dataset: (1) training on the original data using a standard ERM model (Ori.), (2) training on fair data generated by FairData without any fairness-aware objective (Fair), and (3) applying a fairness-aware model (e.g., DiffDP) on the fair data, as shown in Figures 3c and 3f. We can observe:

- In the original setting (Ori.), we observe clear distributional discrepancies between male and female groups. For example, in Figure 3(a) and (d), the male group exhibits flatter, more dispersed prediction densities, while the female group tends to concentrate around low or high probability extremes. This asymmetry reflects strong prediction bias driven by the underlying data distribution.
- After replacing the training data with fair data produced by FairData, but using a vanilla ERM model (Fair), the group distributions become notably more aligned, as seen in Figure 3(c) and (f). Despite no explicit fairness regularizer being used during training, the model trained on FairData's data implicitly learns to produce fairer predictions. This highlights the fairness-inductive property of our data-centric approach.
- Finally, when applying a fairness-aware model on the fair data, the group distributions exhibit near-perfect alignment, as shown



(a) ACS-I (Gender) (b) CelebA-A (Gender)

Figure 4: Impact of the hyperparameter α on the trade-off between error rate and fairness (Δ_{DP}). Increasing α enhances fairness but slightly compromises accuracy, demonstrating the controllable nature of the proposed method.

in Figure 3(b) and (e). This synergy suggests that FairData not only generates inherently fair data, but also complements fairness-aware training objectives when they are applied.

- Across both datasets, we observe that the gap in predictive density between groups shrinks progressively across the three settings: Ori. → Fair → Fair + FairModel. This consistent trend reinforces the robustness and generality of FairData in mitigating distributional bias across different fairness strategies and dataset characteristics.

These results demonstrate that FairData can induce fair decision boundaries even without explicit fairness constraints during training. The learned data distribution enables flexible fairness-accuracy trade-offs and generalizes well to different models, further validating the robustness of our approach.

5.7 Controllable Trade-off via α

We investigate how the hyperparameter α influences the trade-off between prediction accuracy and fairness. By varying α within $(0, 1)$, we adjust the relative strength of the fairness objective in our optimization formulation. As shown in Figure 4, increasing α consistently improves fairness, measured by a reduction in demographic parity violation Δ_{DP}, while slightly increasing the prediction error on both ACS-I and CelebA-A datasets.

Smooth Trade-off Behavior. Notably, on ACS-I, Δ_{DP} drops significantly from over 4.0 to nearly 0.0 as α increases, while the error rate only rises moderately from 16.2% to 18.0%. A similar trend is observed on CelebA-A, where the error rate increases from 15.3% to 24.5% as Δ_{DP} steadily declines from 19.4 to 0.6. These results validate that FairData enables a smooth and monotonic control over the fairness-accuracy trade-off.

Interpretation of the Trade-off. This behavior stems from the dual-objective nature of our framework: a higher α places more emphasis on minimizing fairness violations during the synthetic data learning process, even if it comes at the cost of deviating from the original predictive loss landscape. In contrast, lower values of α prioritize fidelity to the original task, treating fairness as a secondary concern. The ability to interpolate between these two extremes allows practitioners to select a balance point that suits deployment requirements.

Controllability and Stability. The monotonic trends across both datasets further demonstrate the stability and controllability of our formulation. This robustness suggests that the learned fair data respond predictably to variations in α, making the method easy to tune in practice. Importantly, the trade-off curve is relatively flat near $\alpha = 0.5$, indicating a practical operating point where substantial fairness gains are achieved with minimal loss in accuracy.

Task-specific Dynamics. Additionally, we observe dataset-specific dynamics in the impact of α: CelebA-A shows a steeper slope in both fairness improvement and accuracy degradation, implying greater sensitivity to α than ACS-I. This can be attributed to differences in dataset complexity and feature-label correlations, highlighting the importance of context-aware hyperparameter tuning. For instance, CelebA-A's visual modality introduces more latent correlations with sensitive attributes, making fairness interventions more impactful but also more disruptive to accuracy.

In summary, our analysis reveals that FairData provides a controllable mechanism for navigating fairness-utility trade-offs, supporting flexible deployment in fairness-critical applications. It also emphasizes the interpretability and transparency of the optimization process, aligning with broader goals in trustworthy AI.

6 Conclusion and Future Work

We propose FairData, a data-centric framework that improves group fairness by synthesizing a fair training dataset, without modifying downstream model architectures. Experiments on three tabular datasets and one image dataset show that FairData can reduce group disparities while maintaining competitive predictive performance across multiple model families. A key benefit is reusability: once generated, the dataset can be applied to different downstream models, and we observe that the fairness improvements often transfer across architectures (e.g., from MLPs to tree-based or deep models). Overall, FairData provides a practical alternative to model-centric debiasing by shifting the optimization target from model parameters to training data.

As future work, we plan to extend FairData in several directions. First, we will incorporate task-specific constraints or domain knowledge to guide fair data synthesis. Second, we will explore compositional generation to disentangle different bias factors and mitigate them in a more targeted manner. Third, we will study how to integrate FairData with active learning or continual learning pipelines to better support evolving data streams. Finally, it would be interesting to examine whether fairness-aware dataset optimization can benefit a broader set of downstream settings, such as ranking and recommendation systems, multimodal classification and perception pipelines, and retrieval-based or retrieval-augmented applications [51, 55, 57, 58], where fairness considerations are also important.

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Ethical Considerations

FairData aims to improve group fairness by synthesizing fair training datasets. However, fairness is inherently context-dependent, and any single metric may fail to capture all relevant societal considerations. In addition, synthetic data could be misused (e.g., by selectively curating or manipulating samples) to give a misleading impression of fairness. We therefore recommend reporting experimental settings transparently, evaluating multiple fairness metrics when appropriate, and validating the resulting models in the target domain prior to deployment.

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